

MERN Full Stack Final Assessment (FA)

SET D — Leave & Attendance Workflow Management System

API Response & Frontend Evaluator Contract

1. Purpose of This Document

This document defines the:

- expected API response formats
- status codes
- pagination structures
- authentication response structures
- analytics response formats
- frontend evaluator conventions
- Playwright compatibility requirements

for the:

Leave & Attendance Workflow Management System — Set D

Students must strictly follow:

- route names
- response structures
- status codes
- data-testid conventions
- window.appState structure

mentioned in this document.

Failure to follow these conventions may affect automated evaluation.

2. Common Backend Response Standards

Success Response

Status Code

200 / 201

Expected Structure

```
{
  "success": true,
  "message": "Operation successful",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```



Expected Output

Error Response

Status Codes

Scenario	Status
Validation Error	400
Unauthorized	401
Forbidden	403
Not Found	404
Server Error	500

Expected Structure

```
{
  "success": false,
  "message": "Error message"
}
```

Pagination Response Format

Expected Structure

```
{
  "success": true,
  "message": "Data fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

3. MongoDB & Authentication Routes

POST /sync

Status Code

200

Auth Required

Yes



Expected Output

Allowed Roles

admin

hr_manager

Expected Response

```
{
  "success": true,
  "message": "Dataset synchronized successfully",
  "data": {
    "totalFetched": 120,
    "inserted": 92,
    "duplicates": 11,
    "rejected": 17
  }
}
```

Values are dynamic based on dataset.

GET /health

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Database connected successfully",
  "data": {
    "database": "connected",
    "documentCount": 92
  }
}
```

POST /auth/register

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Employee registered successfully",
  "data": {
    "_id": "string",
    "employeeId": "EMP101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
  }
}
```



Expected Output

```
{
  "role": "employee",
  "createdAt": "ISO_DATE"
}
```

POST /auth/login

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Login successful",
  "token": "JWT_TOKEN",
  "data": {
    "_id": "string",
    "employeeId": "EMP101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "hr_manager"
  }
}
```

GET /auth/me

Status Code

200

Auth Required

Yes

Expected Response

```
{
  "success": true,
  "message": "Authenticated employee fetched successfully",
  "data": {
    "_id": "string",
    "employeeId": "EMP101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "employee"
  }
}
```



4. Employee APIs

GET /employees

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Employees fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "employeeId": "EMP101",
      "name": "Dinesh",
      "email": "dinesh@test.com",
      "department": "HR",
      "status": "active"
    }
  ]
}
```

GET /employees/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Employee fetched successfully",
  "data": {
    "_id": "string",
    "employeeId": "EMP101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "employee",
    "department": "HR",
    "designation": "Trainer",
    "leaveBalance": 12,
  }
}
```

Expected Output

```
{
  "status": "active"
}
```

5. Department APIs

POST /departments

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Department created successfully",
  "data": {
    "_id": "string",
    "departmentId": "DEPT101",
    "name": "HR",
    "employeeCount": 12,
    "status": "active",
    "createdAt": "ISO_DATE"
  }
}
```

GET /departments

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Departments fetched successfully",
  "data": [
    {
      "_id": "string",
      "departmentId": "DEPT101",
      "name": "HR",
      "status": "active"
    }
  ]
}
```

GET /departments/:id

Status Code



Expected Output

200

Expected Response

```
{
  "success": true,
  "message": "Department fetched successfully",
  "data": {
    "_id": "string",
    "departmentId": "DEPT101",
    "name": "HR",
    "manager": {},
    "employeeCount": 12,
    "status": "active"
  }
}
```

PATCH /departments/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Department updated successfully",
  "data": {
    "_id": "string",
    "departmentId": "DEPT101",
    "status": "active",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /departments/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Department deleted successfully"
}
```

6. Attendance APIs

POST /attendance



Expected Output

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Attendance marked successfully",
  "data": {
    "_id": "string",
    "attendanceId": "ATT101",
    "employee": {},
    "date": "2026-05-20",
    "status": "present",
    "workingHours": 8,
    "createdAt": "ISO_DATE"
  }
}
```

GET /attendance

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Attendance fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "attendanceId": "ATT101",
      "status": "present",
      "workingHours": 8
    }
  ]
}
```

GET /attendance/:id

Status Code



Expected Output

200

Expected Response

```
{
  "success": true,
  "message": "Attendance fetched successfully",
  "data": {
    "_id": "string",
    "attendanceId": "ATT101",
    "employee": {},
    "date": "2026-05-20",
    "checkIn": "09:00",
    "checkOut": "17:00",
    "status": "present",
    "workingHours": 8
  }
}
```

PATCH /attendance/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Attendance updated successfully",
  "data": {
    "_id": "string",
    "attendanceId": "ATT101",
    "status": "present",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /attendance/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Attendance deleted successfully"
}
```



Expected Output

7. Leave APIs

POST /leaves

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Leave request created successfully",
  "data": {
    "_id": "string",
    "leaveId": "LEV101",
    "employee": {},
    "leaveType": "Casual Leave",
    "approvalStatus": "pending",
    "createdAt": "ISO_DATE"
  }
}
```

GET /leaves

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Leave requests fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "leaveId": "LEV101",
      "leaveType": "Casual Leave",
      "approvalStatus": "pending"
    }
  ]
}
```

GET /leaves/:id



Expected Output

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave request fetched successfully",
  "data": {
    "_id": "string",
    "leaveId": "LEV101",
    "employee": {},
    "approvedBy": {},
    "leaveType": "Casual Leave",
    "fromDate": "2026-05-21",
    "toDate": "2026-05-22",
    "reason": "Personal Work",
    "approvalStatus": "pending"
  }
}
```

PATCH /leaves/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave request updated successfully",
  "data": {
    "_id": "string",
    "leaveId": "LEV101",
    "approvalStatus": "approved",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /leaves/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave request deleted successfully"
}
```

8. Leave Approval Workflow APIs

PATCH /leaves/:id/approve

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave approved successfully",
  "data": {
    "leaveId": "LEV101",
    "approvalStatus": "approved",
    "approvedBy": {
      "_id": "string",
      "name": "HR Manager"
    }
  }
}
```

PATCH /leaves/:id/status

Allowed Status

pending

approved

rejected

cancelled

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave status updated successfully",
  "data": {
    "leaveId": "LEV101",
    "approvalStatus": "approved",
    "updatedAt": "ISO_DATE"
  }
}
```

Workflow Validation Rules

- rejected leaves cannot move directly to approved



Expected Output

- cancelled leaves cannot be approved
- employees cannot approve their own leave
- approved leave should reduce leave balance

9. Filtering, Search & Pagination APIs

GET /attendance?status=present

GET /attendance?department=HR

GET /attendance?date=2026-05-01

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Attendance filtered successfully",
  "data": [
    {
      "_id": "string",
      "attendanceId": "ATT101",
      "status": "present"
    }
  ]
}
```

GET /leaves?approvalStatus=approved

GET /leaves?leaveType=Casual

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Leave requests filtered successfully",
  "data": [
    {
      "_id": "string",
      "leaveId": "LEV101",
      "approvalStatus": "approved"
    }
  ]
}
```



Expected Output

GET /employees?page=1&limit=10

GET /employees?search=dinesh

GET /leaves?approvalStatus=pending&leaveType=Sick&page=2&limit=5

Status Code

200

Expected Pagination Response

```
{
  "success": true,
  "message": "Employees fetched successfully",
  "page": 2,
  "limit": 5,
  "total": 50,
  "totalPages": 10,
  "data": [
    {
      "_id": "string",
      "employeeId": "EMP101",
      "name": "Dinesh"
    }
  ]
}
```

10. Analytics APIs

GET /analytics/attendance

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Attendance analytics fetched successfully",
  "data": {
    "totalEmployees": 120,
    "presentEmployees": 96,
    "absentEmployees": 24,
    "attendancePercentage": 80
  }
}
```

Values are dynamic.

GET /analytics/leaves

Status Code



Expected Output

200

Expected Response

```
{
  "success": true,
  "message": "Leave analytics fetched successfully",
  "data": {
    "totalLeaveRequests": 60,
    "approvedLeaves": 40,
    "rejectedLeaves": 8,
    "pendingRequests": 12
  }
}
```

Values are dynamic.

GET /analytics/departments

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Department analytics fetched successfully",
  "data": [
    {
      "department": "HR",
      "attendancePercentage": 88,
      "highestLeaveRequests": 12,
      "activeEmployees": 25,
      "leaveUtilizationPercentage": 65
    }
  ]
}
```

Values are dynamic.

11. Frontend Evaluator Contract

Mandatory React Router Routes

Route	Purpose
/login	Authentication
/dashboard	Analytics dashboard
/employees	Employee management
/departments	Department management



Expected Output

/attendance	Attendance management
/leaves	Leave management
/profile	Current user profile

Mandatory window.appState Structure

Students must expose:

window.appState

Expected Structure

```
window.appState = {  
  authUser,  
  token,  
  employees,  
  departments,  
  attendance,  
  leaves,  
  filters,  
  analytics  
}
```